

Stakeholder Question Catalogue 04

Research Staff

80 research-delivery questions mapped to proportionate sensing, authoritative institutional sources, explicit operation types, evidence quality, uncertainty, permissions and safe answer boundaries.

STAKEHOLDER	Postdoctoral researchers, research associates and fellows, and project-employed researchers using Abacws for sustained project delivery, analysis, experiments, collaboration, specialist computing, dissemination and irregular schedules.
QUESTION RANGE	Focused work and deadlines; authorised experiments and data collection; project collaboration; computing and hybrid work; visitors and dissemination; irregular hours; accessibility, well-being, sustainability, reliability and responsible evidence use.
EVIDENCE MODEL	Proposed, correctly located sensing is combined with authorised spatial, booking, access, BMS, energy, network, AV, equipment, maintenance, accessibility, weather, research-governance and safety data. Every record states the operation, authority, uncertainty and safe answer boundary.

Core principle: Support a research-delivery decision without inspecting research content, inferring productivity, identifying or tracking people, substituting nearby data, bypassing authoritative systems or sounding more certain than the evidence.

1. Catalogue scope and design principles

The catalogue focuses on decisions that postdoctoral researchers, research associates/fellows and project-employed researchers may reasonably want a conversational university building to support during project delivery, analysis, authorised experiments, collaboration, computing, dissemination and irregular schedules.

Proposed capability, not current fact

Every sensor, integration and service is a design requirement unless deployment, room coverage, permissions and operational status are verified. The catalogue does not claim that all capabilities currently exist.

Current building information

The May 2021 floor plan is a historical spatial reference. Current room use, ownership, staff access, bookability, specialist facilities, equipment, lifts, services and HVAC relationships require authoritative verification.

Research-staff decision context

Questions concern project delivery, sustained analysis, authorised experiments, specialist computing, team coordination, visitors and dissemination, deadline peaks, irregular schedules and responsible evidence use - not doctoral development, teaching duties or facilities operations.

Question-to-evidence design

Each answer resolves the decision, place, period and hard constraints; identifies minimum evidence; checks authority, spatial coverage and data quality; performs the correct operation; and states uncertainty and safe failure.

Research-content boundary

The building must not inspect datasets, code, models, drafts, experiments, presentations, conversations, speech or network content, infer productivity or monitor an individual. Only the minimum user-supplied context needed for the request is processed.

Authoritative source precedence

Bookings, access, room purpose, equipment, network, AV, closures, maintenance, lone-working rules, alarms, research governance and emergencies come from owner-controlled systems and competent authorities.

Privacy and security

Use anonymous aggregate occupancy and numerical sound statistics. Enforce research-staff permissions, minimise personal and project data, and withhold restricted operational details.

Honest answerability

Distinguish observation, authoritative lookup, calculation, estimate, forecast, diagnosis and recommendation. Ask for missing constraints, qualify weak evidence and return not assessable when minimum evidence is absent.

2. How to read each question record

Each question is a natural research-staff request followed by the proposed evidence and controls needed for a responsible answer. Readiness describes implementation dependency, not question importance.

RESEARCH STAFF QUESTION	A realistic natural-language request made by a postdoctoral researcher, research associate/fellow or project-employed researcher.
SENSORS AND TELEMETRY	Physically plausible proposed measurements, required spatial placement, timing, calibration/commissioning, quality and privacy constraints.
AUTHORITATIVE SUPPORTING DATA	Owner-controlled room, access, booking, equipment, service, project, accessibility, governance or safety information that sensing cannot replace.
ANALYSIS REQUIRED	The minimum evidence and method, explicitly distinguishing observation, lookup, calculation, estimate, forecast, diagnosis and recommendation.
ANSWER BOUNDARY	What the building may not infer or claim, how missing/conflicting evidence is handled, and when permission, safety or authority requires refusal or escalation.
WHY THIS IS USEFUL	The concrete project-delivery, research-quality, collaboration, accessibility, reliability or well-being decision supported by the answer.

Readiness

R1 - practical Core proposed sensing plus basic authoritative spatial or service data.	R2 - integration-dependent Booking, access, project, IT, AV, lift, BMS, equipment, maintenance or forecasting integration.	R3 - research-grade or restricted Dense sensing, validated comparison/diagnosis, controlled operational data, or stronger privacy, governance and safety controls.
Global rule: use only verified eligible spaces and authorised data; preserve timestamps and room-level coverage; inspect no research, speech or network content; never infer productivity or identity; and return a limited or not-assessable answer when critical evidence is missing.		

3. Worked example: project-delivery day

A responsible answer combines user-supplied activities and hard constraints with verified access, room purpose, equipment, services, room-level evidence, governance and uncertainty; it never treats an empty or quiet space as automatically available, private or safe.

Research staff member: "I will be in Abacws from 9 a.m. to 9 p.m. I need a four-hour analysis block, a hybrid project meeting, a short authorised equipment run and an evening writing period. Can you plan the day and identify where the evidence is too weak to recommend?"

1. Resolve the workday

Resolve fixed times, durations, access, privacy, equipment, service, accessibility, lone-working and latest-departure constraints from the minimum user-supplied information.

4. Perform the correct operations

Use lookup for permissions, observation for current conditions, forecast for future demand, calculation for routes and diagnosis only when time-aligned evidence supports it.

2. Verify authorised candidates

Use current room purpose, research-staff access, bookings, equipment ownership/status, opening hours and closures; do not infer capability or permission from occupancy or the 2021 plan.

5. Optimise with independent backups

Apply access, equipment, privacy, accessibility and safety constraints first; minimise travel; avoid backups sharing the same unverified room, lift, network or equipment dependency.

3. Verify evidence and governance

Require target-room coverage, environmental freshness, aggregate network/AV status, equipment telemetry, data quality and applicable research, security and lone-working rules.

6. Communicate answerability

State the plan, sources, evidence times, confidence, omitted criteria, recheck points and safe-failure conditions; return conditional or not assessable where essentials are missing.

ILLUSTRATIVE ANSWER STRUCTURE

"For the analysis block, [verified space] is the strongest option because access and service requirements are confirmed and quieter conditions are forecast. Use [verified room] for the hybrid meeting; low occupancy alone is not evidence of privacy. The equipment run remains conditional on [authoritative approval/status]. For evening work, follow [official lone-working rule] and recheck at [time]. Confidence is [low, moderate or high] because [coverage and schedule uncertainty]. [Criterion] was omitted because [missing, stale or restricted source]."

1. Focused research work, deadlines and workspace planning

Questions RS-001-RS-002

RS-001

CORE

L3 - forecast + recommendation

R2 - integration-dependent

Research staff question: My funder report is due in two weeks. Can you find several four-hour blocks when a quiet, comfortable staff workspace I'm authorised to use is likely to be available?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: current research-staff space inventory, room purpose and permitted activity, staff access, opening hours, bookings, planned events, route graph and only the deadline and constraints the researcher supplies. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: forecast and recommendation. Filter to verified eligible spaces; use comparable weekdays, future bookings/events and quality-controlled environmental history to forecast occupancy, sound and stability; rank primary and backup blocks with prediction intervals, travel time and a recheck point. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

This is a probability-based planning recommendation, not a guaranteed desk, quiet period or socket. Do not infer productivity or attendance. If future schedules, room-level coverage or comparable history are inadequate, return broad options or not assessable. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Helps a project-employed researcher protect scarce deep-work time before a funder deadline without turning attendance or apparent activity into performance monitoring.

RS-002

CORE

L2 - current recommendation

R2 - integration-dependent

Research staff question: I've got three hours between project meetings. Where can I work quietly now without ending up far from the next meeting?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: meeting location voluntarily supplied by the researcher; current staff access, room purpose, bookings, opening hours, route graph, power provision and current network/service incident feeds. Missing or permission-denied fields remain unknown. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup, current observation and recommendation. Resolve the three-hour interval; filter verified staff-eligible spaces; compare current and recent occupancy, sound, environment and aggregate service status; rank a nearby primary and fallback option. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record. For observation, state the measurement window, aggregation, freshness, valid coverage and excluded channels.

Answer boundary

Current occupancy does not prove that a desk or socket is free, and low sound does not guarantee future quiet. If target-room evidence or authoritative availability is stale, state that and avoid a precise recommendation. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Reduces transition time and preserves a useful block of project work between fixed commitments.

1. Focused research work, deadlines and workspace planning

Questions RS-003-RS-004

RS-003

CORE

L2 - constrained recommendation

R2 - integration-dependent

Research staff question: I have a confidential sponsor call followed by two hours of analysis. Which bookable room is officially suitable for private conversation and practical to use now?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: current room purpose, acoustic/privacy classification, staff access, bookings, opening hours, network/AV service status and route graph. The researcher supplies only the call duration and required level of privacy; no meeting content or participant identity is required. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: lookup and recommendation. Apply hard constraints for verified bookability, permitted use and privacy classification before using current environment or service quality; rank valid candidates and state which privacy evidence is authoritative versus inferred. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Quietness and low occupancy are not proof of acoustic privacy. Do not listen to speech or inspect call traffic. If no room has authoritative privacy classification and verified access, provide bookable-room options without claiming confidentiality. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Protects sponsor confidentiality and preserves a usable analysis block without assuming that an empty room is private or available.

RS-004

ADVANCED

L4 - matched comparison

R3 - research-grade or restricted

Research staff question: Over the past six weeks, which monitored staff workspaces were most consistently quiet and thermally stable on weekday mornings?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: verified current room classifications, staff eligibility, capacities, events/timetable context, façade or served-zone relationships and outdoor weather. Use only spaces with equivalent measurement coverage and retain historical room/sensor changes. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: calculation and matched historical comparison. Define comparable weekday-morning windows; calculate median, variability, exceedance duration, sound percentiles and data coverage; adjust for occupancy and weather; report effect sizes and uncertainty rather than a simple league table. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

Comparison is valid only for equivalently instrumented spaces with enough overlapping data. Do not rank spaces with materially different coverage or use corridor measurements as room evidence; mark affected candidates not assessable. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Helps the researcher make a defensible long-stay workspace choice and reveals persistent environmental differences rather than one-off impressions.

1. Focused research work, deadlines and workspace planning

Questions RS-005-RS-006

RS-005

IMPORTANT

L3 - demand forecast

R2 - integration-dependent

Research staff question: Our reporting deadline is next week and Abacws may be busy. When am I most likely to find a calm, eligible place for sustained writing?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: future bookings, timetable/events, opening hours, staff access and any project dates voluntarily supplied; historical aggregate occupancy and footfall for comparable deadline periods. Use no individual calendar or attendance history without explicit permission. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: forecast and recommendation. Model time-of-day occupancy, footfall and sound from comparable weeks, scheduled activity and current trend; identify low-demand windows and provide prediction intervals, primary/backup times and a same-day recheck trigger. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

Forecasts may fail when attendance or events change unexpectedly. Do not use individual movement histories or infer who will attend. If schedule coverage is incomplete, label confidence low and avoid precise seat claims. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Allows research staff to plan concentrated work around predictable deadline-related demand peaks.

RS-006

CORE

L2 - capability recommendation

R2 - integration-dependent

Research staff question: I need a full day of visual analysis using a large display, docking and reliable power. Which staff workspace is the best verified option?

Sensors and telemetry

Proposed target-room air temperature, RH, direct NDIR CO2, task-area illuminance and numerical LAeq/L10/L90; authoritative socket/circuit status, docking/display inventory and equipment-service health. No screen, application or device content is inspected. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: authoritative room and workstation inventory, display/docking capability, staff entitlement, bookings, opening hours, planned maintenance, power incident feed and route/accessibility data. Device capability must be verified by the asset owner, not inferred from the floor plan. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup, current observation and recommendation. Verify entitlement, asset capability, booking and live service state. Apply display, docking, software, access and duration as hard constraints; then compare current occupancy, sound and environment, rank a primary and equivalent backup, and set a pre-use recheck. Cite source times, state observation freshness and disclose ranking criteria and uncertainty.

Answer boundary

A listed workstation may still be occupied unless an authorised reservation or check-in confirms availability. Power status does not prove that a socket, display or application works. If entitlement, asset inventory, booking or live status is incomplete, list verified capabilities only and make no availability claim.

Why this is useful: Reduces the chance that a full day of analysis is lost because the required display, docking, power or workspace access was assumed rather than verified.

1. Focused research work, deadlines and workspace planning

Questions RS-007-RS-008

RS-007

IMPORTANT

L4 - itinerary optimisation

R2 - integration-dependent

Research staff question: Tomorrow I need a writing block, a hybrid project meeting and a short approved data-collection session. Can you plan a workable sequence with as little travel as possible?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: researcher-supplied times and hard constraints; current staff access, room purpose, bookings, equipment authorisation, route graph, step-free routes, lift/closure status and opening hours. Use minimum schedule detail and do not read calendar content beyond the request. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: itinerary optimisation and recommendation. Verify each activity and candidate; apply access, equipment, privacy and accessibility constraints; forecast room conditions; minimise route cost; return a primary itinerary, independent backups and recheck times. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

The plan is advisory and must not create or change bookings without an authorised action. If any required room, equipment, route or access status is unverified, flag that activity as conditional or not assessable. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Turns a fragmented project-delivery day into a feasible sequence while preserving time, access, privacy and equipment constraints.

RS-008

IMPORTANT

L3 - historical pattern analysis

R2 - integration-dependent

Research staff question: I found this area difficult to use for focused work last month. Were there recurring periods of high noise, crowding or unstable temperature?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: the location and dates voluntarily supplied; verified sensor mapping, room classification, events, maintenance and weather context. No personal attendance or productivity data is needed. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: historical calculation and pattern detection. Check data quality; calculate duration and recurrence of sound, occupancy and temperature deviations against contextual baselines; align with events and maintenance; report repeated patterns and coverage. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

This can describe measured environmental patterns but cannot prove they caused loss of concentration. Numerical sound statistics reveal no conversation content. If the area lacks valid room-level sensing, return contextual evidence only. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Helps the researcher choose a better workspace or submit a focused facilities report without treating distraction as a productivity metric.

1. Focused research work, deadlines and workspace planning

Questions RS-009-RS-010

RS-009

IMPORTANT

L3 - forecast + alert

R2 - integration-dependent

Research staff question: For my booked work session, can you warn me if the room is likely to become crowded, noisy or poorly ventilated and suggest an authorised backup?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: the verified booking, current room eligibility, scheduled events, approved notification channel and an explicit opt-in with threshold and lead-time preferences. Use only the booking reference and destination needed for this alert; do not create a parallel attendance record. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: short-horizon forecast and opt-in notification. Evidence gate: verified booking plus fresh target-room sensing and a declared alert threshold. Forecast occupancy, numerical sound and CO2 from current trend and scheduled activity; notify only when probability exceeds the threshold and include a verified fallback and evidence time. For forecast, use declared comparable history, report a prediction interval and set a recheck time.

Answer boundary

This is a probabilistic convenience alert, not a safety alarm or guarantee that the room will become unsuitable. Do not track the researcher or reveal other occupants. If booking, target-room coverage or notification permission is missing, do not issue a condition-specific warning. If eligibility, booking, target-room coverage or comparable history is missing, return confirmed constraints and broad options only, or mark the recommendation not assessable.

Why this is useful: Gives the researcher time to move or change plans before a booked focused-work block is likely to be lost.

RS-010

IMPORTANT

L2 - comparative recommendation

R2 - integration-dependent

Research staff question: Given today's verified room and service conditions, is it more practical for me to stay in Abacws or switch to remote work for the rest of the afternoon?

Sensors and telemetry

Proposed in-room environmental and numerical acoustic sensing, anonymous occupancy and authoritative network/power/closure incident status for Abacws only. Remote-home conditions are not sensed; the user may supply only the constraints they wish to compare. Evidence quality: use only target-room or validated served-zone channels; report sensor location, observed-at time, aggregation, commissioning/calibration, uptime, gaps, drift and relocations. Occupancy remains anonymous and acoustics remain numerical.

Authoritative supporting data

Authoritative sources: current Abacws room eligibility, bookings, closures, opening hours and network/power/maintenance incidents, plus only the remote-work constraints and fixed commitments the researcher chooses to provide. No home, device or employer-monitoring data. Authority: current owner-controlled room purpose, staff eligibility, access, booking, opening, closure and feature records take precedence over inferred occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup, bounded comparison and recommendation. Verify Abacws eligibility, current room/service incidents and remaining commitments; compare only the remote-work constraints the researcher supplies. Score feasibility and disruption risk, not productivity; identify missing evidence, disclose criteria and uncertainty, and set a recheck time. Cite source owner and effective/retrieval times.

Answer boundary

The service can compare verified Abacws conditions only with remote constraints the researcher volunteers; it cannot observe the home, judge attendance or infer productivity. If either option lacks minimum evidence, return a conditional comparison or not assessable, not a definitive choice.

Why this is useful: Helps the researcher decide where to continue the afternoon using verified building conditions and only the remote constraints they choose to provide.

2. Experiments, data collection, laboratories and specialist equipment

Questions RS-011-RS-012

RS-011

CORE

L2 - readiness synthesis

R3 - research-grade or restricted

Research staff question: Before I move samples or ask colleagues to attend, can you verify the booking and the equipment owner's current readiness status for this afternoon's approved experiment?

Sensors and telemetry

Authoritative equipment-controller or owner status, interlocks and maintenance state; proposed room temperature/RH and activity-specific environmental or extraction telemetry only where required by the approved method. Apparent power draw is not proof that equipment is ready. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: approved room booking, current room purpose/access, equipment owner readiness/interlock/maintenance status, approved procedure or induction requirements and any planned service outage. Sensor readings cannot substitute for owner release. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authoritative lookup with current observations. Confirm booking, access, owner readiness/interlock and maintenance first; then report time-aligned room conditions relevant to the approved method. Return ready, conditional or not ready with source times; do not infer readiness from telemetry alone. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record. For observation, state the measurement window, aggregation, freshness, valid coverage and excluded channels.

Answer boundary

A booking, low power draw or normal room reading does not prove equipment readiness or experimental safety. The building cannot authorise the work. If owner status, approval or method-critical service evidence is absent, return not ready/unknown and the responsible contact. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Prevents wasted setup time and unnecessary movement before an approved experiment while leaving readiness authority with the room and equipment owners.

RS-012

CORE

L3 - stability forecast

R3 - research-grade or restricted

Research staff question: Over the next seven days, when is this authorised space most likely to provide stable temperature, humidity, particle levels and background sound for my measurement?

Sensors and telemetry

Proposed target-room temperature, RH, PM2.5/PM10, numerical LAeq/L10/L90 and, where method-relevant, airflow/pressure and indicative TVOC; authorised equipment and extraction state. Sensor selection must match the approved measurement sensitivity. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: current room and equipment authorisation, bookings, planned maintenance/cleaning, activity schedule, ventilation schedule and outdoor weather forecast. Use only the stability limits supplied by the research protocol; do not inspect the experiment content. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: forecast and recommendation. Define the method-critical variables and tolerance with the researcher, select only correctly instrumented authorised spaces, model mean and variance across comparable periods, include scheduled activity/weather/service state, and return candidate windows with prediction intervals and recheck times. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

A forecast cannot guarantee low variance or scientific validity. Different instruments may require unmeasured controls. If the approved method, target-room coverage, comparable history or scheduled activity is inadequate, return confirmed availability only or not assessable. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Helps schedule a sensitive measurement when the environmental background is predictably stable, while making uncertainty and unmeasured controls explicit.

2. Experiments, data collection, laboratories and specialist equipment

Questions RS-013-RS-014

RS-013

ADVANCED

L4 - event diagnosis

R3 - research-grade or restricted

Research staff question: An instrument reading changed sharply yesterday. Did it coincide with a room-condition change, equipment power event, ventilation change, occupancy rise or door opening?

Sensors and telemetry

Proposed target-room temperature, RH, direct NDIR CO₂, PM_{2.5}/PM₁₀ and activity-appropriate airflow/extraction; authorised equipment power/health, anonymous occupancy and door events. The system receives only a user-supplied event time, not the instrument value. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: researcher-supplied event time; authorised equipment power/health and non-content acquisition markers; room booking, maintenance, door and ventilation state plus local weather if relevant. Preserve the clock source and synchronisation quality for every stream. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: event reconstruction and bounded diagnosis. Use the researcher-supplied event time, align room/equipment/door/occupancy/ventilation streams on one clock, detect changes at plausible lags and rank supported associations. Preserve a contextual timeline even when no cause is defensible. For diagnosis, require aligned clocks, verified mappings and independent evidence; rank alternatives and allow an unresolved result.

Answer boundary

Coincidence is not causation. The building receives no instrument value or research content. If clocks, mapping or independent context streams are weak, provide the confirmed timeline and competing explanations only, not a cause. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Provides reproducible contextual evidence for troubleshooting an unexpected result without exposing the result itself.

RS-014

ADVANCED

L4 - intervention comparison

R3 - research-grade or restricted

Research staff question: Since the equipment was serviced, have room and service conditions been more stable during comparable runs?

Sensors and telemetry

Proposed correctly located temperature, RH, direct NDIR CO₂, PM_{2.5}/PM₁₀ and indicative TVOC only where the authorised activity warrants them; local extraction/airflow status, equipment power/health, anonymous occupancy and door events. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: authoritative maintenance completion time and affected asset; authorised equipment state; room bookings, activity and weather; researcher-supplied run windows with no experimental values. Maintain sensor and asset mapping across any replacement or relocation. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: matched pre/post calculation and comparison. Verify service time, select runs comparable in duration, occupancy, equipment state, room conditions and schedule, calculate stability/variance and interruption rates, and report effect size, uncertainty and residual confounding. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

A before/after change is not proof that maintenance caused it. If operating conditions or sensor configurations differ materially, state the comparison is confounded or not assessable. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Provides defensible evidence on whether servicing improved repeatability rather than relying on a few favourable post-service runs.

2. Experiments, data collection, laboratories and specialist equipment

Questions RS-015-RS-016

RS-015

IMPORTANT

L2 - recovery assessment

R3 - research-grade or restricted

Research staff question: The room was recently cleaned or used for a dusty task. Have the measured conditions settled back to a stable baseline before I start non-hazardous data collection?

Sensors and telemetry

Proposed target-room PM2.5/PM10, temperature, RH, direct NDIR CO2 and activity-appropriate airflow/extraction; cleaning/task event time from an authorised record. Indicative TVOC may show a broad change but cannot identify a chemical or certify safety. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: authoritative cleaning/activity completion time, room access, permitted next use and ventilation status; researcher-supplied non-hazardous protocol limits. Safety or chemical re-entry decisions remain with competent authorities. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: current observation and baseline-recovery calculation. Compare quality-controlled target-room values with a declared pre-event baseline; require a sustained stable window rather than a single reading and show trend, coverage and uncertainty. This does not release a hazardous process. For observation, state the measurement window, aggregation, freshness, valid coverage and excluded channels. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

A stable low-cost sensor trend does not certify chemical safety, contamination clearance or method suitability. If a hazardous activity or approved clearance rule applies, defer to that rule and competent authority; missing baseline or room coverage means not assessable. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Avoids starting sensitive measurements while residual environmental disturbance is still evident.

RS-016

IMPORTANT

L4 - route optimisation

R2 - integration-dependent

Research staff question: I have approved measurements in several spaces today. What order and route will keep each visit inside its time window while minimising travel?

Sensors and telemetry

Proposed room-level environmental sensors where they provide context; authorised door/lift/closure status and anonymous corridor footfall. Sensor data is not needed to prove access; retain current route-state timestamps and do not track the researcher. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: researcher-supplied locations, time windows and accessibility constraints; current staff access, room purpose, bookings, route graph, lift/door status and any experiment-specific authorisation. Use only minimum itinerary information. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: route and schedule optimisation. Verify each destination and time constraint; compute feasible routes; account for expected congestion, lift state and setup duration; return an ordered plan with backup routes and recheck points. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

The plan does not grant access or override bookings. It must not store a movement trace beyond the request. If a destination, route or permission is unverified, mark that leg conditional or omit it. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Reduces travel and setup overhead in multi-location data collection while preserving required timing and accessibility.

2. Experiments, data collection, laboratories and specialist equipment

Questions RS-017-RS-018

RS-017

ADVANCED

L4 - temporal association

R3 - research-grade or restricted

Research staff question: Did repeated door openings or nearby activity plausibly influence yesterday's environmental measurements?

Sensors and telemetry

Proposed correctly located temperature, RH, direct NDIR CO₂, PM2.5/PM10 and indicative TVOC only where the authorised activity warrants them; local extraction/airflow status, equipment power/health, anonymous occupancy and door events. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: researcher-supplied run interval; authorised room booking and door-state data; anonymous aggregate occupancy and nearby activity context. Door and occupancy coverage, false counts and sensor placement must be documented. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: time-lagged comparison and bounded diagnosis. Align door, anonymous occupancy and environmental streams; compare periods with and without repeated openings under similar ventilation/weather; quantify effect size and lag while keeping unmeasured activity and airflow mixing as alternatives. For diagnosis, require aligned clocks, verified mappings and independent evidence; rank alternatives and allow an unresolved result. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability.

Answer boundary

The analysis can identify temporal association, not individual behaviour or definitive causation. Do not identify who entered. If door/occupancy sensing is incomplete or the sensor is not representative of the experiment location, qualify or abstain. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Helps assess a plausible source of measurement disturbance without monitoring individuals.

RS-018

CORE

L3 - service risk estimate

R3 - research-grade or restricted

Research staff question: Before I start a long acquisition, are the approved power, network and ventilation services currently available, and what verified interruption risks remain?

Sensors and telemetry

Authoritative circuit/UPS status, aggregate network health, ventilation/extraction status and equipment controller health; proposed target-room temperature/RH and any approved method-critical environmental stream. Apparent availability is not equipment authorisation. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: current equipment reservation and permission, power/network maintenance, BMS service schedule, room booking and only the planned duration and protocol limits supplied by the researcher. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authoritative lookup, current observation and interruption-risk estimate. Confirm equipment/service entitlement and owner status, then check fresh power/network/ventilation and room evidence. Estimate only bounded risk from recent incidents/maintenance and return a pre-start recheck trigger. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record. For observation, state the measurement window, aggregation, freshness, valid coverage and excluded channels. For estimate, state assumptions, validation scope and an error or uncertainty range.

Answer boundary

Current service health is not a guarantee of continuity or permission to operate. If entitlement, equipment-owner status, telemetry freshness or incident history is weak, provide confirmed status and a recheck/escalation plan rather than a numerical risk claim. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Avoids committing equipment time to a long acquisition when a required service is unavailable, degraded or poorly evidenced.

2. Experiments, data collection, laboratories and specialist equipment

Questions RS-019-RS-020

RS-019

SPECIALIST

L4 - restricted risk synthesis

R3 - research-grade or restricted

Research staff question: I'm considering an unattended overnight acquisition. What formal approvals are still missing, and which verified building or equipment conditions could interrupt it?

Sensors and telemetry

Authoritative equipment controller/interlock, circuit/UPS, network and ventilation/extraction status plus certified alarms exposed through approved interfaces. Environmental research sensors may provide context but cannot approve unattended operation or replace safety controls. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: approved protocol/risk assessment, equipment owner and unattended-operation policy, room/access booking, lone-working requirements, certified alarm/interface status, planned closures and service maintenance. The chatbot cannot create approval. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: policy/approval lookup and conditional readiness assessment. List mandatory approvals and safety controls, verify each through its owner, identify current service/closure risks and return approved, conditional or not authorised. No sensor pattern may waive a missing control. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The chatbot cannot approve unattended operation, override risk assessment, lone-working or alarm rules, or treat environmental IoT as a safety system. Any missing mandatory approval/control means no-go and referral to the responsible authority. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Clarifies whether an overnight run is formally and technically feasible before samples, equipment time or staff effort are committed.

RS-020

ADVANCED

L4 - evidence packaging

R3 - research-grade or restricted

Research staff question: Can you create a time-stamped contextual record of room and equipment conditions during my approved experiment, excluding all research results?

Sensors and telemetry

Proposed approved room-condition streams and authorised equipment status/power markers only; use the experiment start/end times supplied by the researcher. Exclude instrument output, sample identity, protocol content, speech, video and personal data. Evidence quality: use only approved room or equipment telemetry with verified mapping and a common clock; report native cadence, aggregation, calibration/commissioning, maintenance state, uptime, gaps and drift. Do not inspect experimental output.

Authoritative supporting data

Authoritative sources: authoritative experiment booking and room/equipment identifiers voluntarily supplied; current room/asset mapping, equipment service log and research-governance rules for contextual metadata. Exclude experimental values, files and intellectual content. Authority: approved method, room purpose, access, risk controls, equipment owner status, interlocks and maintenance records take precedence over telemetry-derived readiness. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: permission-controlled calculation and evidence packaging. Join only approved room/equipment status to user-supplied start/end times; compute declared summaries with units, cadence, coverage, gaps and clock quality; include provenance and exclude all research output and personal data. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty. For an evidence package, preserve source-to-result lineage and omit unauthorised or restricted content.

Answer boundary

The package must exclude research values, samples, protocol content, speech, images and identities. It is contextual evidence, not proof of experimental validity or safety. Incomplete permission, clock alignment, mapping or coverage requires omission and an explicit unsupported label. If approval, equipment status, room mapping or critical telemetry is missing or conflicting, do not authorise the activity; give the official owner route and mark the analytical conclusion not assessable.

Why this is useful: Creates reproducible contextual metadata for an approved experiment without exposing samples, results or intellectual content.

3. Project-team collaboration, meetings and cross-disciplinary work

Questions RS-021-RS-022

RS-021

CORE

L3 - constrained forecast

R2 - integration-dependent

Research staff question: Eight of us need a hybrid project meeting next week. Which bookable room has confirmed capacity, working AV, dependable connectivity, suitable acoustics and step-free access?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: current room inventory, capacity, permitted use, staff/visitor access, bookings, AV capability and health, accessibility features, lift/closure status and opening hours. Participants provide only required time, size and access needs. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authoritative lookup, forecast and recommendation. Apply booking, capacity, access, AV, accessibility and privacy as hard constraints; verify current service state; forecast occupancy/environment only for valid rooms; rank primary and independently verified backup with trade-offs and a recheck. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

A quiet room is not automatically private or bookable. Network/AV health is not a guarantee. Do not inspect meeting content or participant accounts. If any hard requirement lacks authoritative confirmation, do not recommend that room. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Reduces hybrid-meeting failure and supports equitable participation by treating capacity, accessibility, AV and privacy as verified constraints.

RS-022

IMPORTANT

L4 - recurring optimisation

R2 - integration-dependent

Research staff question: Our cross-disciplinary team meets fortnightly. Which recurring room and time are most practical for commuters and remote participants?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: only team-authorised availability windows, current room bookings, staff access, opening hours, visitor rules, accessibility, route information and permitted external transport feeds. Do not read calendar subjects, infer caring or commuting circumstances, or retain individual schedules beyond the request. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: recurrence optimisation and recommendation. Use only availability patterns supplied or authorised, apply access/booking/AV/accessibility constraints, estimate travel and remote-join feasibility, forecast room demand, and rank recurring options with sensitivity to missing attendance information. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

The result cannot guarantee attendance or privilege those who are already on site. External transport and future bookings may change. If consented availability, accessibility or authoritative room status is incomplete, return feasible ranges rather than a definitive recurring schedule. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Helps recurring collaboration across mixed working patterns without privileging only colleagues who are already on site.

3. Project-team collaboration, meetings and cross-disciplinary work

Questions RS-023-RS-024

RS-023

CORE

L2 - current recommendation

R1 - practical

Research staff question: Four colleagues are free now. Where can we have a short project discussion nearby without using a restricted or already booked space?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: current staff-accessible room inventory, permitted activity, bookings, opening hours, route graph and room capacity. The user supplies only group size and current location; no identity list is required. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: current authorised lookup and recommendation. Verify eligibility and availability; filter by capacity and meeting suitability; compare current occupancy, sound and distance; return a primary and backup option. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

An apparently empty room is not necessarily available, and open collaboration space is not confidential. If booking or access status is unavailable, list only spaces explicitly designated for ad hoc staff use. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Helps quick collaboration while respecting room permissions and other building users.

RS-024

ADVANCED

L4 - matched comparison

R3 - research-grade or restricted

Research staff question: For a two-hour methods meeting, which eligible room has shown the most stable acoustics, ventilation and hybrid-call service in comparable periods?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: verified room types, capacities, staff access, bookings and historical authorised AV/network incident data. Use comparable meeting periods and preserve any equipment or sensor changes. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: matched historical comparison. Restrict to equivalent rooms and meeting windows; calculate sound variability, CO2 trend, service interruptions and data coverage; adjust for occupancy and duration; report effect sizes and uncertainty. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability.

Answer boundary

Past performance does not guarantee the next meeting. Do not inspect call traffic or speech. If rooms lack equivalent sensing/service history, report comparability limits and avoid ranking them. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Helps the team choose a repeatable meeting environment using matched historical evidence rather than reputation or one recent experience.

3. Project-team collaboration, meetings and cross-disciplinary work

Questions RS-025-RS-026

RS-025

IMPORTANT

L3 - scenario forecast

R2 - integration-dependent

Research staff question: We're planning a half-day cross-disciplinary workshop. Which start time is most likely to reduce arrival queues and keep occupancy and ventilation manageable?

Sensors and telemetry

Proposed anonymous entrance, corridor and lift-lobby counts; doorway counts for the verified event room; distributed room CO2 and temperature for a large occupied space; numerical sound only if audibility context is needed. State counter coverage, count error, sampling and missingness; identify no attendee and derive no individual route. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: expected attendance supplied by organisers; current room capacity and booking; event schedule, building opening, visitor/accessibility arrangements, corridor route graph, lift status and historical aggregate arrival patterns. No attendee tracking is required. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: arrival-demand forecast and recommendation. Combine expected attendance, event schedule, entrance/route capacity and comparable anonymous counts; simulate candidate starts, estimate queues and room environmental load, and return a range plus registration/ventilation recheck triggers. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

Forecasts are sensitive to actual attendance and travel disruption. The system cannot direct crowd control or override venue limits. If expected attendance or authoritative room/access data is missing, provide qualitative planning guidance only. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Helps organisers choose a workshop start time that reduces congestion and environmental risk while preserving uncertainty about actual attendance.

RS-026

IMPORTANT

L4 - visitor itinerary

R2 - integration-dependent

Research staff question: A visiting collaborator will spend the day with our project. Can you plan an accessible itinerary for meetings, a demonstration and a quiet online-work block?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: visitor-hosted schedule and stated access needs; current visitor permissions, room bookings, authorised demonstration/equipment access, route graph, step-free routes, lift/closure status, opening hours and public waiting areas. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: itinerary optimisation and recommendation. Verify each activity; apply visitor, equipment, privacy and accessibility constraints; minimise travel; forecast space conditions; provide primary and backup routes/rooms and check-in instructions. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

The system must not infer disability, nationality or security status. It cannot grant visitor or specialist access. If approvals or routes are unverified, mark the affected activity conditional and direct the host to the authoritative process. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Improves a collaborator's visit while reducing logistical burden on the project team.

3. Project-team collaboration, meetings and cross-disciplinary work

Questions RS-027-RS-028

RS-027

IMPORTANT

L4 - space-set optimisation

R2 - integration-dependent

Research staff question: Our team wants a project co-working day with one shared base and two nearby breakout rooms. Which bookable arrangement is feasible without displacing scheduled activity?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: current room purposes, staff access, capacities, bookings, opening hours, route proximity, permitted group use and power/network capability. Team supplies expected size, hours and whether discussion or quiet work dominates. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: constrained space-set optimisation. Filter authorised rooms; find a base plus two nearby valid breakouts; forecast occupancy/environment; minimise route distance and conflict; return a primary configuration and fallback. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

Low occupancy does not permit use of a booked or restricted room. The recommendation cannot reserve spaces unless an authorised booking action is invoked. If no complete set is verified, return partial options rather than invent capacity. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Helps productive project co-location while respecting other users and formal room allocation.

RS-028

CORE

L2 - real-time recommendation

R2 - integration-dependent

Research staff question: Our meeting is running over. Is there an authorised nearby room we can book for one more hour without affecting someone else's booking?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: current meeting location and group size supplied by the user; authoritative bookings, permitted room use, opening hours, access and route graph. Current occupancy can support context but does not override the booking source. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: real-time lookup and recommendation. Identify rooms available for the extension interval, check capacity and permitted activity, compare distance and current conditions, and return a primary and backup option. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The service must not assume an empty space is free or extend a booking without authorisation. If no authoritative availability feed exists, provide only designated drop-in spaces and state the limitation. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Allows collaboration to continue without unintentionally occupying someone else's booked room.

3. Project-team collaboration, meetings and cross-disciplinary work

Questions RS-029-RS-030

RS-029

ADVANCED

L4 - cross-domain diagnosis

R3 - research-grade or restricted

Research staff question: Our last hybrid meeting had poor audio and repeated connection drops. Does the evidence point to the room AV, local network, power, an external endpoint or no defensible diagnosis?

Sensors and telemetry

Authoritative AV device health, room microphone/speaker test status, aggregate AP/link latency and packet-loss probes, circuit/UPS events and numerical room acoustics. No meeting audio, network traffic, account, device or remote-endpoint content is inspected. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: meeting time and room, AV owner diagnostics, aggregate network incident/AP-link health, circuit/UPS events, room maintenance and only a user report of the observed symptom. No meeting recording or participant endpoint logs. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: incident reconstruction and bounded diagnosis. Align AV self-test, aggregate network, power and room-acoustic events; compare room/floor/building scope and external endpoint uncertainty; rank supported fault domains and provide a factual support bundle without causal certainty. For diagnosis, require aligned clocks, verified mappings and independent evidence; rank alternatives and allow an unresolved result.

Answer boundary

Room/AV/network evidence may locate a fault domain but cannot diagnose a participant device or external platform without authorised data. No audio or traffic is inspected. If independent evidence or clocks are inadequate, return the confirmed incident chronology only. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Helps the project team route a hybrid-meeting fault to the right support domain without inspecting call content or claiming a cause that the evidence cannot support.

RS-030

ADVANCED

L4 - availability comparison

R3 - research-grade or restricted

Research staff question: Across this term, which shared staff spaces were most dependable for collaboration without recurring crowding, noise or service problems?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: room and route sensing is contextual only; report target-room placement, observed-at time, coverage and outages. Occupancy remains anonymous, sound remains numerical, and AV/network evidence is aggregate service health only.

Authoritative supporting data

Authoritative sources: verified room purpose, staff access, bookings, capacity and historical aggregate use. Use anonymous occupancy only; do not construct individual attendance or team movement histories. Authority: current booking, capacity, visitor, accessibility, room-purpose, AV and closure records take precedence over occupancy-derived availability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: historical calculation and matched comparison. Define collaboration-eligible spaces; calculate availability, occupancy distribution, sound percentiles and data coverage for comparable periods; report reliability and uncertainty. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

This measures space-level patterns, not who used the spaces or whether collaboration was productive. If booking and occupancy sources disagree, report both and do not infer unauthorised use. If booking, capacity, accessibility, visitor, AV or room evidence is stale or missing, provide confirmed facts and a conditional fallback, not a guarantee.

Why this is useful: Helps evidence-based planning of recurring team collaboration without monitoring individual researchers.

4. Computing, data workflows, remote and hybrid research

Questions RS-031-RS-032

RS-031

CORE

L2 - service recommendation

R2 - integration-dependent

Research staff question: I need a three-hour interactive remote-compute session. Which authorised workspace has the strongest verified network and power conditions without inspecting my work?

Sensors and telemetry

Authoritative aggregate wired/Wi-Fi service health, latency and packet-loss probes, circuit/UPS status and target-room temperature; anonymous occupancy only for workspace context. No packets, endpoint identity, code, data, prompts or compute-job content are inspected. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: current staff access, room purpose, bookings, opening hours, authorised network/power service status and any secure-computing requirements supplied by the researcher. Use institutional service metadata, not personal device or job content. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised current-status lookup and recommendation. Evidence gate: verified staff eligibility, current room status and aggregate network/power telemetry for the relevant service area. Compare service health, recent incidents, occupancy and room conditions; rank a primary and fallback and show the evidence time and service scope. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Network probes cannot guarantee application performance, and no traffic or remote-session content may be inspected. If room-level network evidence is unavailable, report building/service status only and avoid a precise room ranking. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Supports selection of a suitable workspace for interactive remote computing without inspecting traffic, code, files or model activity.

RS-032

IMPORTANT

L4 - incident diagnosis

R3 - research-grade or restricted

Research staff question: My remote analysis slowed suddenly at 14:20. Is there evidence of a wider network or power incident, a room-local problem, or nothing the building can diagnose?

Sensors and telemetry

Authoritative aggregate AP/link health and incident records, circuit/UPS events and target-room temperature; the researcher supplies only time and location. No packets, account logs, device identity, files, code, data or remote-service content are inspected. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: researcher-supplied event time and room; authorised network incident/AP-link summaries, circuit/UPS events, room/cooling state and building maintenance. No access to the analysis, remote host, files, device or account logs. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: event reconstruction and bounded diagnosis. Align authorised network, power and room/cooling records around 14:20; compare room, floor and building scope; classify the evidence as wider service, room-local, endpoint-unknown or not assessable, with source times and clock quality. For diagnosis, require aligned clocks, verified mappings and independent evidence; rank alternatives and allow an unresolved result.

Answer boundary

The system cannot diagnose the researcher's device or remote service without authorised evidence. It must not inspect network content. If source clocks or coverage are inadequate, return the confirmed incident timeline only. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Helps decide whether to contact building/IT support or investigate the endpoint, while keeping the analysis and device content private.

4. Computing, data workflows, remote and hybrid research

Questions RS-033-RS-034

RS-033

IMPORTANT

L3 - transfer forecast

R2 - integration-dependent

Research staff question: I must upload a large authorised dataset before a deadline. Which approved place and time are likely to be most reliable, and what secure fallback is available?

Sensors and telemetry

Authoritative aggregate network service health and approved secure-transfer service status, circuit/UPS events and target-room temperature. No file names, contents, destinations, credentials, endpoint identity or transfer payload are inspected. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: approved transfer service and data-governance route, network owner status, workspace access/booking, opening/closure, circuit/UPS and maintenance. The researcher provides only deadline, approximate transfer size and constraints needed for planning. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup, throughput estimate and recommendation. Confirm permitted transfer route and workspace/service availability, estimate completion from privacy-preserving aggregate throughput history and declared size, report error range and recommend a secure fallback plus latest safe start time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record. For estimate, state assumptions, validation scope and an error or uncertainty range.

Answer boundary

The system cannot inspect the dataset, guarantee throughput or bypass approved transfer/governance routes. If secure service entitlement, destination policy or usable aggregate history is missing, do not recommend an improvised transfer; return the official route and uncertainty. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Helps deadline-critical transfer planning while preserving security, privacy and honest uncertainty about throughput.

RS-034

CORE

L2 - resource recommendation

R3 - research-grade or restricted

Research staff question: I need a managed workstation for a long compute job. Which resource that I am entitled to use is available, adequately cooled and clear of planned maintenance?

Sensors and telemetry

Authoritative workstation/compute inventory, entitlement, scheduler/availability, maintenance and aggregate health; asset inlet/room temperature, cooling status and circuit/UPS state where owner-exposed. No job, process, model, code or dataset content is inspected. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: compute/workstation entitlement and booking/scheduler, asset owner availability/maintenance, room access/opening, power/cooling incidents and planned outages. Do not infer entitlement or readiness from occupancy or power draw. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup and recommendation. Filter to entitled resources; verify reservation and service state; assess current thermal/power margin and planned works; rank a primary and backup machine or service. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The building cannot inspect code, datasets or job parameters and cannot promise job completion. If equipment ownership, entitlement or live state is not available, direct the researcher to the official compute service. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Prevents a long compute task being assigned to a resource that is unavailable, unentitled, under maintenance or inadequately cooled.

4. Computing, data workflows, remote and hybrid research

Questions RS-035-RS-036

RS-035

CORE

L2 - impact lookup

R2 - integration-dependent

Research staff question: Are any planned building, network or compute outages likely to disrupt the rooms and services my work package needs this week?

Sensors and telemetry

Authorised aggregate wired/Wi-Fi availability, latency/packet-loss probes, circuit/UPS/PDU status and equipment inlet or room temperature where exposed; anonymous occupancy only for workspace context. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: authoritative building works, IT service notices, compute maintenance, room closures, access changes and only the locations/services the researcher selects. Use owner-maintained notices with effective dates and audience permissions. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup and impact mapping. Resolve the user-specified services and times; map notices to rooms/assets and project activities; identify conflicts and suggest verified alternatives or recheck dates. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The system must not read project plans beyond user-supplied constraints. Absence of a notice is not a guarantee of uninterrupted service. If asset-to-room mapping is uncertain, state the possible rather than definite impact. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Allows research staff to move tasks before planned outages threaten a deliverable or experiment.

RS-036

CORE

L2 - hybrid-room recommendation

R2 - integration-dependent

Research staff question: I have a remote collaborator meeting this afternoon. Which bookable room best combines confirmed privacy classification, healthy AV, reliable network service and low background noise?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: current room privacy classification, staff access, bookings, AV capability/health, network incident state, opening hours and route/accessibility data. Use no participant identity or meeting content. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: current authorised lookup and recommendation. Apply privacy and booking constraints; verify AV/network state; compare sound and occupancy; rank primary and backup rooms and provide a pre-call test time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Low sound and occupancy do not prove privacy. Service health does not guarantee a remote platform. Do not record speech or inspect call traffic. If no room has verified privacy and service capability, state that clearly. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Improves remote collaboration quality without exposing the discussion or personal device data.

4. Computing, data workflows, remote and hybrid research

Questions RS-037-RS-038

RS-037

ADVANCED

L4 - service comparison

R3 - research-grade or restricted

Research staff question: During my usual analysis hours, which staff workspaces had the most reliable aggregate network service last month?

Sensors and telemetry

Authoritative aggregate AP/link availability, latency and packet-loss probes scoped to candidate workspaces, with owner-defined privacy-preserving aggregation; target-room power and thermal context only where relevant. No user, device, packet or application content. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: verified staff-space classifications, access, room/AP mapping, authorised aggregate network metrics and maintenance events. The researcher supplies usual time windows; no account, device or browsing history is required. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: historical calculation and matched comparison. Restrict to comparable instrumented spaces and time windows; calculate outage rate, latency/packet-loss distributions, coverage and confidence; adjust for known maintenance. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

Aggregate network metrics may not represent every application or device. Do not infer that the researcher was present. If AP-to-room mapping or data coverage is weak, report service-area results rather than a room ranking. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Helps the researcher make a repeatable choice of workspace for network-dependent analysis.

RS-038

SPECIALIST

L4 - policy + risk synthesis

R3 - research-grade or restricted

Research staff question: Am I allowed to leave an approved data transfer running after I go, and which power, network or closing conditions could stop it?

Sensors and telemetry

Authorised aggregate wired/Wi-Fi availability, latency/packet-loss probes, circuit/UPS/PDU status and equipment inlet or room temperature where exposed; anonymous occupancy only for workspace context. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: official data-handling and unattended-transfer policy, managed-device entitlement, session timeout, room access/closing rules, maintenance, power/UPS and network service status. The system needs no file names or contents. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: policy lookup and bounded interruption-risk synthesis. Evidence gate: official data-handling/unattended-transfer rule, managed-device entitlement and authorised service status. State whether the action is permitted, required safeguards, scheduled closing or maintenance constraints and known interruption risks; provide the official IT route for uncertainty. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The conversational service cannot grant an exception, bypass data-handling rules or keep a session alive. It does not inspect filenames, contents or destinations. If policy, managed-device ownership or service state is unclear, return no permission decision and direct the researcher to the official service. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Prevents avoidable transfer failure or policy breach when a legitimate task may continue after the researcher leaves.

4. Computing, data workflows, remote and hybrid research

Questions RS-039-RS-040

RS-039

ADVANCED

L4 - equipment diagnosis

R3 - research-grade or restricted

Research staff question: A managed workstation stopped during a job. Do the authorised records support overheating, power loss, room-cooling failure, a wider service event or only an unresolved endpoint problem?

Sensors and telemetry

Authoritative workstation/compute health, scheduler event, circuit/UPS status, room/cooling telemetry and aggregate network incident status. The system receives a job-stop time only and inspects no process, code, model, data or output. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: researcher-supplied event time and managed asset ID; authoritative equipment health, power/UPS, cooling/BMS and maintenance events. Exclude job content, code, datasets and personal endpoint logs. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: incident reconstruction and bounded diagnosis. Align the user-supplied stop time with workstation health, scheduler, power, cooling and network events; test room/floor/building scope and rank hypotheses. Leave the endpoint cause unresolved when content-free telemetry cannot distinguish it. For diagnosis, require aligned clocks, verified mappings and independent evidence; rank alternatives and allow an unresolved result.

Answer boundary

Content-free telemetry cannot always distinguish endpoint, application or remote-service failure. Do not inspect the job or user activity. If asset mapping, clocks or service records are incomplete, report only confirmed events and leave the cause unresolved. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Helps route a failure to the appropriate support domain and preserves a reproducible incident record.

RS-040

IMPORTANT

L3 - incident evidence package

R3 - research-grade or restricted

Research staff question: Can you give me a precise, time-stamped account of any building power or network interruption during yesterday's approved computation?

Sensors and telemetry

Authoritative circuit/UPS and aggregate network incident records with service scope, event and ingestion times; target-room temperature only as context. No user/device attribution, packets, files, account or computation content. Evidence quality: use aggregate room, AP, circuit or service telemetry with verified scope and clock alignment; report coverage, maintenance windows, gaps and uncertainty. Inspect no traffic, endpoints, accounts, files, code or models.

Authoritative supporting data

Authoritative sources: researcher-supplied room/service and interval; authoritative network and power event logs, UPS/PDU state, building maintenance and clock-quality records. No compute-job content or user identity is required. Authority: IT entitlement, managed-resource status, incident and maintenance records, and verified room/AP/circuit mappings take precedence over inferred service quality. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup and evidence packaging. Reconcile events across sources; normalise timestamps; identify confirmed interruption start/end, scope and gaps; report source precedence and uncertainty. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record. For an evidence package, preserve source-to-result lineage and omit unauthorised or restricted content.

Answer boundary

The timeline cannot prove that the interruption caused a scientific result or job failure. Restricted infrastructure details must be redacted. If sources conflict, present the conflict rather than choose silently. If entitlement, room/service mapping or aggregate service evidence is incomplete, report confirmed status or incidents only and leave endpoint cause unresolved.

Why this is useful: Provides an auditable contextual timeline for a disrupted computation without attributing activity to people or inspecting the job.

5. Seminars, visitors, demonstrations, dissemination and project events

Questions RS-041-RS-042

RS-041

CORE

L3 - event forecast

R2 - integration-dependent

Research staff question: We expect about 40 people at a research seminar. Which verified room and start time are most likely to support audibility, ventilation, AV and step-free access?

Sensors and telemetry

Proposed distributed target-room temperature, RH and direct NDIR CO2, reconciled anonymous doorway counts and numerical LAeq/L10/L90; authoritative AV/network/power, lift and door status. Sensor positions must represent the occupied zone, not a corridor. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: organiser-supplied attendance/duration; current room purpose, capacity, booking, visitor access, AV capability/health, accessibility, lift/closure, opening and event schedule. Historic floor-plan labels are not current room confirmation. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: capacity-constrained forecast and recommendation. Verify room purpose, booking, capacity, visitor/accessibility and AV first; use comparable events to forecast arrival, occupancy, CO2, temperature and audibility context; rank room/start-time options with prediction intervals and recheck times. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

Capacity, visitor access, accessibility and AV capability are hard constraints. A forecast cannot guarantee attendance, audibility or air quality. If any hard source or distributed room coverage is missing, exclude the room or give a conditional option with recheck. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Reduces the risk that a research seminar fails because the chosen room, time or access arrangement was unsuitable for the expected audience.

RS-042

IMPORTANT

L4 - accessible route optimisation

R2 - integration-dependent

Research staff question: A visiting research group will attend demonstrations on several floors. Can you plan a step-free route that respects access restrictions and avoids predicted congestion?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: host-supplied destinations and times; current visitor permissions, room bookings, approved demonstration spaces, route graph, step-free routes, lift/door/closure status and check-in requirements. No visitor profiling is needed. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: time-dependent route optimisation. Verify each destination and access rule; remove restricted or unavailable paths; forecast congestion from events/footfall; return primary and fallback routes with recheck points. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

The route cannot grant access or replace host responsibility. It must not track visitors. During an emergency, official instructions and responders take precedence over routine route optimisation. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Makes a multi-stop visit easier to manage and more inclusive for guests unfamiliar with the building.

5. Seminars, visitors, demonstrations, dissemination and project events

Questions RS-043-RS-044

RS-043

IMPORTANT

L3 - queue forecast

R3 - research-grade or restricted

Research staff question: Our project event starts at 5 p.m. Where are arrival queues most likely, and how early should registration open?

Sensors and telemetry

Proposed entrance, corridor and lift footfall counters; anonymous room occupancy; door/lift state and room-level environment in waiting areas. Use aggregate counts with documented coverage and count error; identify no visitor. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: expected attendance and arrival profile, event schedule, authorised entrance plan, registration capacity, route graph, accessibility arrangements and other building events. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: queue forecast and timing recommendation. Use expected arrival profile, route/entrance capacity and comparable anonymous counts; simulate opening times, report queue-length/time ranges and uncertainty, and set an operational recheck before the event. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

This is organiser planning support, not live crowd control, capacity approval or emergency guidance. It must not identify or track visitors. If expected attendance, entrance layout or counter coverage is inadequate, return qualitative risks and the responsible planning contact. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Helps organisers open registration early enough to control queues without pretending that forecast attendance is certain.

RS-044

CORE

L2 - event-space recommendation

R2 - integration-dependent

Research staff question: We need a bookable area for posters and live demonstrations. Which current option has verified capacity, visitor access, power, network service and suitable lighting?

Sensors and telemetry

Proposed target-area illuminance, temperature, RH, direct NDIR CO2, numerical acoustics and reconciled anonymous counts; authoritative circuit/socket, network, AV, lift and door status. Demonstration equipment readiness comes from its owner, not power draw alone. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: event booking, current area purpose/capacity, visitor and research-staff access, fire/egress constraints, power/socket, network/AV and accessibility inventories, equipment-owner readiness and closures. Floor-plan furniture is not a capacity approval. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: capability lookup and recommendation. Apply event, capacity, access and equipment constraints; verify service state; compare illumination, expected crowding and route accessibility; rank valid candidates. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The service cannot approve equipment, alter egress or infer safe load from socket count. If any capacity, access, power or safety requirement is unverified, do not recommend the space for the demonstration. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Reduces late discovery that a dissemination space lacks the practical or authorised infrastructure required.

5. Seminars, visitors, demonstrations, dissemination and project events

Questions RS-045-RS-046

RS-045

IMPORTANT

L2 - rehearsal recommendation

R2 - integration-dependent

Research staff question: I need to rehearse a conference talk with slides and remote questions. Which room is the strongest verified option for a realistic practice session?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: current room booking, staff access, AV inventory/health, network status, acoustic/privacy classification, opening hours and route/accessibility data. The service needs no presentation content. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: lookup and recommendation. Filter to verified bookable rooms with required AV; compare sound, room size, connectivity and current conditions; rank primary and backup rooms and suggest an authorised pre-test. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The building must not record speech or inspect slides. Availability and AV health do not guarantee platform compatibility. If equipment capability is unknown, list rooms with verified basic presentation facilities only. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Provides a practical rehearsal environment that resembles the technical demands of the real event.

RS-046

CORE

L3 - event readiness

R3 - research-grade or restricted

Research staff question: Before attendees arrive, is the booked room ready for a hybrid research event, or is the verified backup the safer option?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: authoritative booking, capacity, AV and streaming capability, network service status, staff/visitor access, accessibility, event support arrangements and backup booking. Use no stream content or attendee identity. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: readiness synthesis and recommendation. Verify all hard requirements; run authorised aggregate AV/network checks; assess current room environment and known incidents; compare primary and backup and state unresolved risks. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

The service cannot certify broadcast quality or override event-support judgement. If any required AV, network, capacity, accessibility or access source is missing, report the room as conditionally ready or not assessable. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Helps the researcher make a timely venue decision before speakers and attendees arrive.

5. Seminars, visitors, demonstrations, dissemination and project events

Questions RS-047-RS-048

RS-047

IMPORTANT

L4 - contingency planning

R2 - integration-dependent

Research staff question: If the room, lift or AV service becomes unavailable on the event day, what authorised fallback sequence should our project team follow?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: primary and backup bookings, room/lift/AV incident state, visitor and accessibility plan, capacity, route graph, opening hours and organiser-approved communication and support channels. A fallback is valid only if its access and critical dependencies are independently verified. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: contingency lookup and recommendation. Evidence gate: verified independent fallback resources and current routes. For room, lift and AV failure scenarios, define triggers, accessible alternatives, recheck times, actions requiring organiser authorisation and official contact points; expose shared dependencies. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The service must not move the event, issue public messages or override safety instructions without an authorised action. If no independent fallback or accessible route is verified, state that clearly and escalate to the responsible organiser or service owner. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Gives the project team an authorised contingency that can be acted on quickly instead of improvising after visitors arrive.

RS-048

ADVANCED

L4 - event diagnosis

R3 - research-grade or restricted

Research staff question: After yesterday's seminar, do the records support crowding, ventilation, AV or network faults, or a combination, as the main contributors to the reported problems?

Sensors and telemetry

Proposed distributed room temperature, RH, direct NDIR CO2, anonymous entrance counts and numerical acoustics; authoritative AV/network/power and ventilation events. Use no attendee identities, speech, meeting content, device logs or research material. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: seminar interval and room supplied by the organiser; authorised booking, attendance estimate, AV/network incident, BMS and maintenance records. Exclude recordings, speech, slides and participant identities. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: event reconstruction and bounded diagnosis. Align occupancy, environmental, ventilation, AV, network and power events for the seminar; compare spatial patterns and plausible lags, quantify supported associations and retain multiple contributors when evidence does not separate them. For diagnosis, require aligned clocks, verified mappings and independent evidence; rank alternatives and allow an unresolved result.

Answer boundary

The result identifies supported contributors, not blame or proven causation. It must not inspect attendees, speech or research content. If event time, distributed room coverage or AV/network clocks are inadequate, provide a limited chronology and mark diagnosis not assessable. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Turns mixed attendee reports into a time-aligned, privacy-preserving account of supported contributors and unresolved uncertainty.

5. Seminars, visitors, demonstrations, dissemination and project events

Questions RS-049-RS-050

RS-049

IMPORTANT

L2 - privacy-bounded recommendation

R2 - integration-dependent

Research staff question: I have a short media interview about our project. Which bookable visitor-appropriate room is quiet and officially classified for the privacy we need?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: current room purpose, visitor permission, staff booking rights, privacy/acoustic classification, bookings, opening hours, route/accessibility and any communications policy supplied by the institution. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup and recommendation. Apply visitor, privacy and booking constraints; compare current numerical sound and occupancy; rank valid rooms and provide a backup. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Low measured sound does not prove privacy. The service must not record or analyse speech. If visitor use or privacy classification is unverified, recommend only officially designated interview/meeting spaces. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Helps clear public communication while protecting confidentiality and respecting room rules.

RS-050

ADVANCED

L3 - event evidence package

R3 - research-grade or restricted

Research staff question: Can you produce a concise, permission-controlled evidence summary of room conditions and service interruptions during our public demonstration, excluding attendees and research content?

Sensors and telemetry

Proposed target-room environmental and numerical acoustic summaries, reconciled anonymous counts and authorised AV/network/power incident records for the event interval. Exclude attendee identity, speech, images, research content and restricted topology. Evidence quality: use room-level environmental/acoustic sensing, anonymous flow counts and aggregate AV/network health with current timestamps, calibration/commissioning and coverage. Do not identify attendees or capture speech.

Authoritative supporting data

Authoritative sources: authoritative event booking, room, schedule, AV/network/power incidents and organiser-approved event metadata. Exclude visitor identities, recordings, demonstration content and research data. Authority: event booking, capacity, visitor/accessibility, AV, security and closure records take precedence over predicted demand or sensed occupancy. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: permission-controlled event summary. Join only approved environmental/service streams to the verified event interval, calculate declared summaries and incident chronology, report quality/provenance and redact identities, speech, research content and restricted system detail.

Answer boundary

The summary does not evaluate scientific success or attendee satisfaction. Restricted operational details must be minimised. If a relevant source is absent or stale, mark that component unavailable rather than infer it. If event booking, capacity, visitor/accessibility, AV or route evidence is incomplete, return a conditional plan with recheck points and official contacts.

Why this is useful: Provides an accountable event record for reporting and lessons learned without identifying attendees or disclosing research content.

6. Late, early, weekend, irregular and lone-working patterns

Questions RS-051-RS-052

RS-051

CORE

L1 - authoritative lookup

R2 - integration-dependent

Research staff question: I need to arrive at 6 a.m. on Saturday. Is my access valid, which entrance should I use and which essential services are officially scheduled to be available?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: current building hours, personal research-staff access entitlement through the authorised system, permitted entrance, planned closures, lift/service schedule, security guidance and the user-supplied destination. Do not disclose access-control topology. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: authorised lookup. Resolve the requested time; check access entitlement, entrance and service status; report confirmed facts, effective times, restrictions and official contact routes. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Occupancy or an unlocked door does not prove permission. The service cannot grant entry or reveal sensitive access details. If entitlement or entrance status is unavailable, direct the researcher to the official access/security channel. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Prevents an unnecessary early journey and clarifies which services can actually be relied upon.

RS-052

CORE

L4 - restricted recommendation

R3 - research-grade or restricted

Research staff question: I need to work late on a deadline. Which workspace and exit route are permitted under today's lone-working rules, and which services are scheduled to remain active?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: current lone-working procedure, researcher access, building hours, permitted after-hours spaces, check-in channel, security/support arrangements, bookings, closures, service schedules and verified normal or step-free exit routes. Anonymous occupancy does not determine permission or safety. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: policy lookup and constrained recommendation. Evidence gate: current rule, verified access window, check-in/support arrangement and exit route. Apply mandatory conditions first; then identify eligible workspaces and active services, provide a departure trigger and name the official escalation channel. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Authorised entry does not remove lone-working obligations or prove that support is available. Do not reveal who else is present or infer safety from occupancy. If policy, check-in, route or support status is unverified, make no late-work recommendation. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Helps policy-compliant late work without treating building access or anonymous occupancy as evidence that lone working is safe.

6. Late, early, weekend, irregular and lone-working patterns

Questions RS-053-RS-054

RS-053

SPECIALIST

L4 - policy + readiness

R3 - research-grade or restricted

Research staff question: Can this approved experiment run unattended overnight, and which official conditions must be met before I start it?

Sensors and telemetry

Authorised equipment state, power/UPS, environmental conditions, leak detection and owner-controlled alarms where relevant. Sensing provides context only and cannot approve unattended research. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: official unattended-operation and lone-working policy, risk assessment, equipment/room authorisation, access, booking, maintenance and emergency contacts. Use only the researcher-supplied experiment identifier and duration. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: authoritative policy lookup and readiness checklist. Enumerate mandatory approvals and service prerequisites; verify current/scheduled status; return unmet conditions and no-go cases before any risk estimate. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

No sensor reading can substitute for permission, risk assessment or certified safety systems. If any required approval or status is absent, the answer must be not authorised/not assessable. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Prevents unsafe or unauthorised overnight operation and clarifies the formal route to approval.

RS-054

CORE

L2 - late-work recommendation

R2 - integration-dependent

Research staff question: I'll be in Abacws until 10 p.m. Which permitted workspace still has confirmed lighting, network service and a practical route to the authorised exit?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: current research-staff access, room purpose, bookings, opening/closing times, authorised exit routes, lift/door status, service schedule and lone-working requirements. Use no individual tracking. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: current lookup and recommendation. Filter to permitted late-use spaces; verify network/power/lighting and route status; compare current conditions and distance; return primary and backup spaces plus a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

This is routine planning, not emergency evacuation guidance. The system cannot guarantee continued quiet or services. If access, exit or lone-working requirements are unverified, do not recommend the space. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Helps the researcher make a practical late-working decision while keeping formal access and departure requirements visible.

6. Late, early, weekend, irregular and lone-working patterns

Questions RS-055-RS-056

RS-055

SPECIALIST

L2 - authorised notification

R3 - research-grade or restricted

Research staff question: Can the service remind me to use the official lone-working check-in and send any missed-confirmation alert only through the authorised workflow?

Sensors and telemetry

Only the official lone-working workflow may hold the user identity, check-in time, missed-confirmation event and escalation state. The conversational layer may request/remind through an authorised API but must not derive compliance from occupancy or location tracking. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: current lone-working procedure, authorised check-in/escalation platform, the user-created session and official security contact. The conversational service stores only the minimum reminder state permitted by policy. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: authorised reminder and escalation handoff. Evidence gate: explicit opt-in and live integration with the official lone-working workflow. Schedule only the approved reminder, transmit no extra location or research detail, and relay a missed-confirmation state to the official service without inventing an independent alarm path.

Answer boundary

The conversational service cannot approve lone work, silently track presence or replace the official escalation platform. If the authorised workflow is unavailable or policy does not permit integration, provide the official instructions only and create no shadow reminder system. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Supports policy-compliant lone-working reminders while keeping escalation inside the official workflow and minimising personal data.

RS-056

ADVANCED

L4 - service comparison

R3 - research-grade or restricted

Research staff question: For the services I depend on, were weekends less reliable than comparable weekday periods?

Sensors and telemetry

Authorised aggregate network, lift, BMS and power service events; ROOM_ENV for equivalently monitored rooms. Retain service scope, timestamps, maintenance exclusions, coverage and source authority. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: verified opening schedules, planned maintenance, room classifications and comparable use/occupancy context. Use no individual weekend attendance. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: matched reliability comparison. Define the services and valid service window, compare weekends with equivalent weekday periods after controlling for planned maintenance, opening and demand, calculate incident/downtime rates with confidence intervals, and flag sparse-event uncertainty. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability.

Answer boundary

Past weekend reliability does not guarantee a future visit, and some reductions may be planned policy rather than failure. Do not generalise from one room or service. If logging differs by period, report descriptive evidence without a reliability ranking. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Shows whether weekend work carries a genuine service-reliability penalty after planned closures and lower demand are considered.

6. Late, early, weekend, irregular and lone-working patterns

Questions RS-057-RS-058

RS-057

CORE

L2 - pre-journey check

R2 - integration-dependent

Research staff question: I have a long commute. Before I leave home, can you verify that my destination, lift and essential services are expected to be available at my arrival time?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: researcher-supplied arrival time and destination; official access, booking, room/equipment status, lift, closures, planned maintenance and opening hours. Optional public-transport status comes from an authoritative external feed. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: future-status lookup and short-horizon forecast. Verify scheduled availability and current incidents; identify risks to arrival; provide a recheck time and verified alternative if available. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Future status may change, and transport feeds are external. The system cannot guarantee access or equipment. If the destination or entitlement is not verified, state that before the researcher travels. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Avoids a wasted commute when a critical building or service dependency is unavailable.

RS-058

IMPORTANT

L2 - route-time calculation

R2 - integration-dependent

Research staff question: My last train is earlier tonight. What is the latest reasonable departure time using a verified normal or step-free route, given current lift or corridor disruptions?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: researcher-supplied destination/transport preference; authoritative building closing and route status, lift/door closures, routine security guidance and external public-transport timetable/service alerts. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: authorised lookup and route-time calculation. Resolve the transport deadline; compute normal accessible travel time with a safety buffer; account for verified route disruptions; provide a leave-by time and fallback route. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

This is routine travel planning, not emergency guidance. External services can change. Do not infer the researcher's location or track departure; if route or transport data are stale, state the uncertainty and recommend a larger buffer. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Helps reliable commuting after irregular research hours without unnecessary personal monitoring.

6. Late, early, weekend, irregular and lone-working patterns

Questions RS-059-RS-060

RS-059

CORE

L2 - early-run readiness

R3 - research-grade or restricted

Research staff question: I have an early-morning equipment run. Will the authorised room, ventilation, power and equipment service be ready at the scheduled start?

Sensors and telemetry

Proposed correctly located temperature, RH, direct NDIR CO2, PM2.5/PM10 and indicative TVOC only where the authorised activity warrants them; local extraction/airflow status, equipment power/health, anonymous occupancy and door events. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: current booking, staff/equipment permission, building access, service start schedules, planned maintenance, approved procedure and owner-maintained equipment status. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: future-status lookup and readiness synthesis. Verify all authorisations and scheduled services; compare expected conditions with the user-supplied operating envelope; report confirmed ready, conditional and unknown prerequisites. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The service cannot approve the run or certify safety. If any required room, equipment, ventilation, power or access source is missing, the answer must remain conditional or not ready. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Reduces the risk of arriving early for a run that cannot start because building services are not yet active.

RS-060

SPECIALIST

L1 - safety information

R3 - research-grade or restricted

Research staff question: If something goes wrong while I'm working after hours, what is the current official procedure and which authoritative channel should I use?

Sensors and telemetry

No research sensor is required. Use the current official emergency, security, first-aid and lone-working communication channels and, where authorised, certified alarm/status feeds. Experimental IoT must not generate emergency instructions. Evidence quality: access, opening, alarm and lone-working status must come from authoritative systems; sensing is contextual only. Report effective time, freshness, service scope and outages, and do not track the individual.

Authoritative supporting data

Authoritative sources: current University/building emergency, security, first-aid and lone-working procedures, official contact channels and active certified alerts where the user is authorised to see them. Retrieved version and effective time must be shown. Authority: access entitlement, building hours, closures, lone-working/check-in rules, official travel or support information and emergency communications take precedence over sensor context. Preserve effective and retrieval times.

Analysis required

Operation: authoritative procedure lookup. Identify the relevant official channel for the stated situation, present the current procedure and contact details, and suppress speculative diagnosis or route optimisation unless an authorised incident instruction explicitly provides it. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The service must never invent emergency advice, delay contacting emergency services or expose security-sensitive detail. Certified alarms and current official instructions take precedence. If the authoritative source is unavailable, direct the user to the published emergency channel immediately. If current access, lone-working, closure or emergency information is unavailable or conflicting, do not recommend attendance; direct the user to the official channel.

Why this is useful: Ensures research staff receive current authoritative instructions rather than a sensor-generated or LLM-invented emergency response.

7. Accessibility, inclusion, well-being, safety and responsible working

Questions RS-061-RS-062

RS-061

CORE

L2 - accessibility recommendation

R2 - integration-dependent

Research staff question: I need a step-free, quiet workspace with an adjustable desk and an accessible toilet nearby. Which currently eligible staff space has all of those features confirmed?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: user-supplied functional requirements; current staff access/bookings, room purpose, verified adjustable-desk and accessible-toilet inventory, step-free route graph, lift/door/closure status and opening hours. Do not request a diagnosis. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup and recommendation. Treat accessibility and equipment requirements as hard constraints; verify routes and room capability; compare current sound/environment and distance; return primary and backup options. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

Do not infer disability or store requirements beyond the request. An empty room is not automatically available. If any accessibility feature or route is unverified, state that and do not recommend the candidate as meeting the requirement. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Helps independent workspace selection while requiring only the functional needs the researcher chooses to disclose.

RS-062

IMPORTANT

L3 - dynamic route recommendation

R2 - integration-dependent

Research staff question: Can you route me to my meeting on a step-free path while avoiding the busiest and noisiest areas as far as current evidence allows?

Sensors and telemetry

Anonymous corridor/lift footfall and numerical sound statistics with documented coverage; authorised lift/door/closure status. No cameras, identities, device tracking or speech recording. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: destination and access needs supplied by the researcher; current route graph, step-free constraints, staff access and temporary closures. Current room purpose comes from authoritative records. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: time-dependent route recommendation. Generate valid routes; remove inaccessible/restricted paths; score by current or short-horizon crowding and sound; return a primary and fallback route with confidence. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

Footfall and sound coverage may be incomplete and conditions can change quickly. Do not infer a medical or sensory condition. During an emergency, official route instructions take precedence. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Reduces avoidable sensory load and congestion during routine movement through the building.

7. Accessibility, inclusion, well-being, safety and responsible working

Questions RS-063-RS-064

RS-063

CORE

L2 - inclusive meeting recommendation

R2 - integration-dependent

Research staff question: A visiting collaborator needs step-free access and caption-capable hybrid AV. Which bookable room and verified route meet both requirements?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, illuminance and numerical sound statistics; anonymous occupancy and footfall; authorised aggregate network/AV health and circuit or socket-service status where relevant. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: visitor and staff access, room booking, capacity, verified caption/display/audio capability, accessible route, lift/door status and any reasonable-adjustment information voluntarily supplied. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: constrained lookup and recommendation. Apply visitor, accessibility and AV requirements first; verify route and service state; compare valid rooms and provide backup arrangements. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The service must not infer health or disability, and it cannot guarantee third-party caption quality. If assistive capability or route status is not authoritative, mark the room unsuitable or unverified rather than assume. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Helps inclusive collaboration and avoids discovering an access or AV barrier after the visitor arrives.

RS-064

CORE

L2 - contextual condition check

R2 - integration-dependent

Research staff question: I've developed a headache in this room. What are the current measured conditions, and is anything clearly unusual compared with this room's reliable baseline?

Sensors and telemetry

Proposed target-room air temperature, RH, direct NDIR CO2, PM2.5/PM10, numerical acoustics and, where installed, airflow; authoritative room/ventilation status. These observations describe the room, not the cause of a symptom. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: user-supplied room and time; current room/ventilation status, recent maintenance/cleaning events where permitted, target-room sensor metadata and the official facilities, occupational-health or emergency reporting routes. No health record is accessed. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: current observation and contextual comparison. Report target-room values, freshness, coverage and deviation from a reliable room baseline under similar time/weather/occupancy; identify only measured anomalies and provide the official reporting route. Do not diagnose the headache. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability. For observation, state the measurement window, aggregation, freshness, valid coverage and excluded channels.

Answer boundary

Room observations cannot diagnose or dismiss a symptom, and normal readings do not prove the space is safe for the individual. For severe or urgent symptoms follow official medical/emergency guidance. Missing/stale room evidence means report the concern, not a normality claim. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Helps the researcher decide whether to move, ventilate or report an environmental concern without turning building data into medical advice.

7. Accessibility, inclusion, well-being, safety and responsible working

Questions RS-065-RS-066

RS-065

IMPORTANT

L2 - schedule recommendation

R2 - integration-dependent

Research staff question: I have a long analysis day with fixed meetings. Can you suggest practical break points and workspace changes without judging my health or productivity?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: researcher-supplied fixed times, preferred break frequency and accessibility needs; current room access, bookings, opening hours, routes and available rest areas. The system needs no productivity, health or research-content data. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: schedule calculation and optional recommendation. Evidence gate: only the fixed commitments and preferences the researcher volunteers, plus verified room/access status. Calculate feasible break and transition windows, avoid unnecessary travel and offer options; do not score focus, health, output or adherence. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

The service cannot prescribe breaks, diagnose fatigue or infer productivity from movement, device use or room occupancy. The researcher remains in control and may ignore the suggestion. If schedule or accessible-space information is incomplete, offer general intervals rather than a personalised plan. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Helps sustain responsible working during an intensive day without turning the building into a personal-performance monitor.

RS-066

CORE

L2 - privacy-minimised recommendation

R3 - research-grade or restricted

Research staff question: I have a temporary workplace adjustment. Which authorised workspace best meets the requirements I choose to share today?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: only the functional requirements the researcher chooses to share, plus authorised reasonable-adjustment/support route where applicable, current room features, access, booking, opening, closures and relevant service/environment status. Do not request or retain diagnosis. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: privacy-minimised constrained recommendation. Translate user-stated functional needs into hard constraints; verify candidates and routes; compare current environment; return primary and backup spaces without storing diagnosis. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

Accessibility and agreed adjustment requirements are hard constraints, not preference weights. Do not infer disability or health status, disclose the request or recommend an unverified feature. If no eligible space is confirmed, return not found and the official support route. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Allows a researcher to obtain practical building support while retaining control over sensitive information.

7. Accessibility, inclusion, well-being, safety and responsible working

Questions RS-067-RS-068

RS-067

CORE

L2 - condition + reporting

R3 - research-grade or restricted

Research staff question: There is a strong odour in this workspace. What relevant measurements are available, what can they support, and how should I report the concern?

Sensors and telemetry

Proposed target-room temperature, RH, direct NDIR CO2, PM2.5/PM10, indicative TVOC and ventilation/extraction status where installed. Generic VOC/gas responses cannot identify a substance, exposure or safety level. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: user-supplied room/time and odour report; current room/ventilation status, authorised maintenance/cleaning events and official facilities/safety reporting routes. No conclusion is drawn from absent incident tickets. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: current observation plus authoritative reporting guidance. Evidence gate: fresh correctly located environmental streams and current room/ventilation context. Report what is and is not measured, whether any value is unusual relative to a reliable baseline, and the official facilities or safety reporting route; make no chemical identification. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record. For observation, state the measurement window, aggregation, freshness, valid coverage and excluded channels.

Answer boundary

Generic TVOC/gas sensors cannot identify the source, exposure or safety. A normal reading does not invalidate the odour report. For acute symptoms or suspected hazard, leave the area and use official safety/emergency guidance; otherwise use the authorised reporting route. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Helps the researcher report a potential environmental issue with accurate location, time and available evidence.

RS-068

CORE

L2 - accessibility contingency

R2 - integration-dependent

Research staff question: The lift on my usual route is unavailable. What current step-free alternative or suitable relocated workspace is officially verified?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: authoritative lift fault/status, current accessible route graph, door/closure status, staff access, bookings, room purpose and the researcher-supplied destination or workspace requirements. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: authorised lookup and recommendation. Remove routes depending on the failed lift; compute verified step-free alternatives; if the destination is inaccessible, identify an eligible substitute space with comparable requirements. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The service cannot assume a route is accessible because it appears connected on the historical plan. If no current verified step-free alternative exists, state that and provide the official assistance contact. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Reduces disruption and supports continued work when a critical accessibility asset fails.

7. Accessibility, inclusion, well-being, safety and responsible working

Questions RS-069-RS-070

RS-069

SPECIALIST

L4 - restricted contingency

R3 - research-grade or restricted

Research staff question: I may need to work alone late and require a step-free route. Which check-in, workspace, exit and contingency arrangements officially apply to me?

Sensors and telemetry

Owner-controlled door, lift and building-service status with event timestamps; anonymous aggregate occupancy and corridor footfall for route context; room-level environmental sensing only for comfort. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: official lone-working and reasonable-adjustment procedures, researcher access, authorised check-in service, current room eligibility, accessible routes, lift/door status, opening hours and security/emergency contacts. Use only user-authorised adjustment data. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: policy lookup and constrained contingency recommendation. Evidence gate: current lone-working rule, authorised check-in/support arrangement, eligible workspace and verified accessible exit. Enumerate mandatory actions first, then provide a primary and fallback route/workspace with recheck triggers and official contacts. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback. For lookup, cite the source owner, effective time and retrieval time; do not extrapolate beyond the record.

Answer boundary

The chatbot cannot approve lone work, infer disability or replace the official safety and accessibility services. If any mandatory check-in, accessible exit, support arrangement or access state is unverified, return no late-work recommendation and direct the researcher to the responsible authority. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Makes accessibility and lone-working requirements explicit before an after-hours visit rather than after a route or support failure.

RS-070

ADVANCED

L4 - aggregate equity analysis

R3 - research-grade or restricted

Research staff question: Using aggregate evidence only, are some staff-workspace types consistently less accessible or environmentally suitable than others?

Sensors and telemetry

Proposed in-room temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous room occupancy and corridor footfall only for route context. Evidence quality: accessibility features and closures come from authoritative records; sensors support current route or room context only. Report freshness, route/target-room coverage and missing data, and process only requirements the user chooses to share.

Authoritative supporting data

Authoritative sources: current space classifications, staff eligibility, accessibility inventory, route/lift history, bookings and maintenance. No individual demographic, attendance or health data is required. Authority: current accessibility inventories, lift/door status, access, booking, agreed support routes and user-shared functional requirements take precedence over inferred suitability. Preserve source owner, effective time, retrieval time and permission scope.

Analysis required

Operation: aggregate accessibility/environment comparison. Group only verified space types, apply a declared coverage/comparability threshold, compare route/feature availability and environmental stability with uncertainty, and report structural gaps without linking results to people or adjustments. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability.

Answer boundary

The analysis cannot infer who was disadvantaged, measure individual well-being or use sensitive adjustment data. Unequal instrumentation can create false differences. Any space group below the comparability or coverage threshold must be marked not assessable. If a required accessibility feature, lift/route status or room-level evidence is unverified, do not assume suitability; return verified alternatives and the official support route.

Why this is useful: Supports inclusive workspace review using aggregate evidence without profiling staff or exposing workplace adjustments.

8. Space/service reliability, sustainability and responsible evidence use

Questions RS-071-RS-072

RS-071

ADVANCED

L4 - reliability analysis

R3 - research-grade or restricted

Research staff question: Which staff-accessible workspaces show recurring environmental or service disruptions that could threaten sustained project delivery?

Sensors and telemetry

Sensor catalogue and telemetry-quality metadata: observed property, unit, room, mounting, served zone, sampling and aggregation, commissioning/calibration, maintenance, uptime, gaps, drift, relocation and source agreement; the relevant in-room observations and authorised service state. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: current space purpose/access, bookings, maintenance/work orders, service incidents and room/asset relationships. Use aggregate patterns only and preserve changes in room use or instrumentation. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: recurrence and reliability analysis. Define service/environment events consistently, require comparable target-room coverage, join authorised incidents/maintenance, calculate frequency, duration and affected periods, and rank project-delivery risk with uncertainty and evidence gaps.

Answer boundary

This identifies space-level reliability patterns, not staff performance. Incident absence may reflect missing monitoring. Do not rank spaces with materially different coverage without explicit qualification. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Helps a project team avoid repeatedly unreliable spaces and prepare evidence-based contingencies for sustained delivery.

RS-072

ADVANCED

L4 - intervention evaluation

R3 - research-grade or restricted

Research staff question: After the energy-saving control change, did environmental conditions or service reliability in research spaces change during comparable occupied periods?

Sensors and telemetry

Authorised building, floor, HVAC, lighting and selected plug-load meters; anonymous occupancy; in-room temperature, CO2 and illuminance; BMS airflow, setpoint and equipment state only for verified serving zones; outdoor weather. Preserve meter boundaries, interval, clock alignment, commissioning, coverage, gaps and asset-zone mapping. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: authoritative intervention date and affected system, verified room/asset relationships, maintenance, bookings/occupancy context and complaints in aggregate. Preserve sensor and control changes across the study window. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: intervention evaluation. Verify intervention date/scope and zone mapping; use matched periods or interrupted time series where defensible, normalise for weather, occupancy and schedule, compare energy, CO2, temperature and service reliability, and report effect sizes, uncertainty and sensitivity.

Answer boundary

A post-intervention association is not proof of causation, comfort or compliance. The service cannot change controls or judge individual experience. If the intervention, comparator, asset-zone mapping or pre/post evidence is inadequate, return not assessable rather than a success/failure claim. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Tests whether an efficiency intervention may have shifted costs onto research conditions or service reliability without overclaiming causality.

8. Space/service reliability, sustainability and responsible evidence use

Questions RS-073-RS-074

RS-073

IMPORTANT

L3 - sustainability calculation

R3 - research-grade or restricted

Research staff question: During our project's usual hours, where is there credible evidence that heating, cooling, ventilation or lighting continues while spaces are unoccupied?

Sensors and telemetry

Authorised building, floor, HVAC, lighting and selected plug-load meters; anonymous occupancy; in-room temperature, CO2 and illuminance; BMS airflow, setpoint and equipment state only for verified serving zones; outdoor weather. Preserve meter boundaries, interval, clock alignment, commissioning, coverage, gaps and asset-zone mapping. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: verified space purpose/access, operating schedules, room bookings and planned events. Anonymous occupancy must have stated count uncertainty; no individual attendance data is used. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Why this is useful: Identifies plausible efficiency opportunities during project hours without treating anonymous occupancy or shared-system behaviour as proof of waste.

Analysis required

Operation: calculation and opportunity screening. Require verified vacancy, bounded equipment/HVAC/lighting telemetry and known service relationships; align states, exclude preconditioning, cleaning, safety and shared-zone demand, and estimate apparently avoidable duration/energy as a range. For calculation, show inputs, units and the rule or objective used, and carry input uncertainty.

Answer boundary

This is an investigation lead, not proof of waste or authority to change controls. Anonymous occupancy may undercount and shared systems may serve other spaces. If vacancy or asset allocation is uncertain, report only an apparent mismatch with its uncertainty. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

RS-074

CORE

L3 - evidence-quality assessment

R2 - integration-dependent

Research staff question: How strong is the evidence behind the building's current answer about air quality in this room?

Sensors and telemetry

Current sensor catalogue and target-room observations: property, unit, location/mounting, served zone, cadence/aggregation, commissioning/calibration, maintenance, uptime, gaps, drift, relocation and cross-sensor agreement; authorised ventilation state. Corridor values remain labelled context. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: current sensor inventory, room/served-zone mapping, commissioning/calibration and maintenance records, recent uptime/gaps/relocation and authorised ventilation state. The interpretation criterion and its owner must be stated explicitly. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Why this is useful: Shows whether an air-quality answer is well supported, partially supported or not assessable before the researcher relies on it.

Analysis required

Operation: evidence-quality and answerability assessment. Check current inventory, target-room/served-zone mapping, property coverage, freshness, cadence, commissioning/calibration, uptime, gaps, drift, relocation and source agreement; classify each air-quality dimension as directly observed, inferred, absent or unreliable. For answerability, classify each evidence obligation as met, partial, conflicting or missing and ask only essential clarification.

Answer boundary

CO2 is mainly a ventilation indicator, not complete air quality; indicative TVOC cannot identify compounds, and corridor data is not room evidence. Stale, uncommissioned, poorly mapped or incomplete sensing permits only a limited statement or not assessable, never healthy/safe language. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

8. Space/service reliability, sustainability and responsible evidence use

Questions RS-075-RS-076

RS-075

ADVANCED

L4 - dataset suitability

R3 - research-grade or restricted

Research staff question: Which building sensor streams are defensible as contextual metadata for my study, and which limitations must I disclose?

Sensors and telemetry

Authorised sensor streams plus complete quality/provenance metadata: observed property, unit, room/served zone, cadence/aggregation, clock, commissioning/calibration, maintenance, uptime, gaps, drift, relocation, transformations and licence/retention constraints. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: approved research-data access, source-owner permission, catalogue/provenance, calibration/maintenance, spatial mapping, retention/licence and transformation records for the requested period. Availability does not imply research permission. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: authorised data discovery and suitability assessment. Match variables, room/zone, cadence and period to the proposed contextual use; calculate valid coverage, gaps, drift and clock alignment; document transformations, selection bias, licence and provenance; issue a suitability card per stream.

Answer boundary

Availability does not imply permission or methodological fitness. The assistant must not inspect the study or expose restricted operations. Streams with inadequate provenance, calibration, mapping, licence or coverage are contextual-only or unsuitable, not silently repaired. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Helps responsible secondary use of building data and transparent disclosure of provenance, coverage and bias.

RS-076

ADVANCED

L4 - reproducibility comparison

R3 - research-grade or restricted

Research staff question: For repeated measurements at the required time of day, which of these two authorised rooms provides the more reproducible environmental background?

Sensors and telemetry

Proposed correctly located temperature, RH, direct NDIR CO₂, PM_{2.5}/PM₁₀ and indicative TVOC only where the authorised activity warrants them; local extraction/airflow status, equipment power/health, anonymous occupancy and door events. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: current room purpose/access, bookings, equipment constraints, comparable activity and weather context. The researcher supplies the required period and variables; room equivalence must be verified rather than assumed. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: matched historical comparison and recommendation. Verify equivalent authorised use and sensor placement, compare within-room variance, autocorrelation and disturbance frequency at the required time, adjust for occupancy/weather/service state, and report effect size, uncertainty and residual confounding. For comparison, predefine matched windows, confounders and a valid-data threshold; report effect size and residual non-comparability. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

The answer compares environmental background, not scientific validity or equipment performance. If room use, equipment, ventilation regime, sensing or coverage differs materially, state that the comparison is confounded and make no definitive room recommendation. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Helps repeatable contextual measurements by comparing environmental variability, not by claiming one room guarantees better science.

8. Space/service reliability, sustainability and responsible evidence use

Questions RS-077-RS-078

RS-077

IMPORTANT

L3 - recurrence synthesis

R3 - research-grade or restricted

Research staff question: I've reported the same room problem several times. Does the available evidence show a recurring issue that is reasonable to escalate?

Sensors and telemetry

Sensor catalogue and telemetry-quality metadata: observed property, unit, room, mounting, served zone, sampling and aggregation, commissioning/calibration, maintenance, uptime, gaps, drift, relocation and source agreement; the relevant in-room observations and authorised service state. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: researcher-supplied ticket references if authorised; owner-maintained work orders, room access/purpose, maintenance actions and planned works. Do not expose other reporters or complaint text. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: recurrence synthesis and escalation support. Link only authorised ticket references, target-room evidence and maintenance events; build a chronology, calculate frequency/duration and post-intervention persistence, and produce a factual escalation bundle with gaps and uncertainty.

Answer boundary

The building cannot assign blame, set work-order priority or disclose other reporters. A missing ticket or sensor gap is not evidence that the issue did not occur. If linkage or room coverage is incomplete, provide a limited chronology and mark recurrence uncertain. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Provides a concise, evidence-based recurrence record that can be escalated without assigning blame or revealing other reporters.

RS-078

CORE

L4 - answerability analysis

R2 - integration-dependent

Research staff question: What evidence is missing, stale or too weak for the building to answer my question about this room responsibly?

Sensors and telemetry

Sensor catalogue and telemetry-quality metadata: observed property, unit, room, mounting, served zone, sampling and aggregation, commissioning/calibration, maintenance, uptime, gaps, drift, relocation and source agreement; the relevant in-room observations and authorised service state. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: the natural-language question, current data catalogue, room/asset graph, source permissions, calibration/maintenance and applicable institutional criteria. Process only the minimum request context. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: question decomposition and answerability assessment. Resolve variable, room/zone, time, comparator and requested operation; test authority, permission, spatial coverage, cadence, calibration, completeness and conflicts; classify each obligation as met, partial, conflicting or missing and ask only essential clarification. For answerability, classify each evidence obligation as met, partial, conflicting or missing and ask only essential clarification.

Answer boundary

Do not fill evidence gaps with nearby sensors, generic building knowledge or a fluent LLM guess. Explain missing or conflicting evidence without exposing restricted infrastructure. If an essential obligation is unmet, return not assessable and state what would be required. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Makes the building explain why it can or cannot answer, enabling the researcher to request the right evidence rather than accept false certainty.

8. Space/service reliability, sustainability and responsible evidence use

Questions RS-079-RS-080

RS-079

ADVANCED

L4 - resilience forecast

R3 - research-grade or restricted

Research staff question: Our deadline week depends on several rooms and services. What are the main availability risks, and which backups avoid the same failure points?

Sensors and telemetry

Authorised aggregate room, network, lift, power and equipment health; relevant room environment for verified critical spaces. Preserve event scope, source and maintenance exclusions. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: researcher-supplied critical activities and times; current bookings, staff access, asset/service relationships, planned maintenance, historic incidents and verified backup resources. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: multi-service risk forecast and contingency recommendation. Map researcher-supplied critical activities to verified room/service dependencies, estimate bounded risks from current status and adequate incident history, identify common-cause failures, rank independent backups and set switch/recheck triggers. For forecast, use declared comparable history, report a prediction interval and set a recheck time. For recommendation, apply permission, safety, accessibility and capability constraints before ranking; state criteria, uncertainty and a verified fallback.

Answer boundary

Rare-event probabilities may be unstable and do not guarantee continuity. Do not expose security-sensitive topology or claim a backup is independent without verified dependency mapping. If history or backup status is weak, provide a qualitative risk register rather than numerical certainty. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Protects time-critical project delivery by exposing shared dependencies and genuinely independent backups before deadline week.

RS-080

ADVANCED

L4 - governed evidence package

R3 - research-grade or restricted

Research staff question: Can you produce a transparent, permission-controlled evidence package for our project report that includes relevant building conditions but excludes research content and restricted operational details?

Sensors and telemetry

Only the authorised contextual sensor/service streams and quality metadata within the approved project scope; preserve units, room/asset mapping, cadence, aggregation, clock, commissioning/calibration, maintenance, uptime, gaps, drift, transformations and source lineage. Exclude research content and personal data. Evidence quality: verify room/asset mapping, meter or sensor hierarchy, cadence, common clock, calibration/commissioning, uptime, gaps, drift, maintenance state, transformations and provenance. Separate direct measurements from estimates or inferred states.

Authoritative supporting data

Authoritative sources: project-approved interval and locations, data-access permission, source-owner restrictions, room/asset mapping, sensor/service provenance, retention/licence and sponsor or research-governance requirements. Exclude project files, intellectual content, identities and restricted operational detail. Authority: governance, permission, current metadata, data-owner/licence terms, maintenance and source records determine whether evidence may be used. Analytics cannot repair missing authority or provenance; preserve effective and retrieval times.

Analysis required

Operation: permission-controlled, provenance-preserving evidence packaging. Validate approved scope and source permissions; calculate only authorised contextual summaries; include units, method, timestamps, cadence, coverage, calibration, uncertainty, transformations and source-to-result lineage; redact restricted detail. For an evidence package, preserve source-to-result lineage and omit unauthorised or restricted content.

Answer boundary

The package is contextual evidence, not proof of scientific validity, safety, compliance or project success. It must exclude research content, personal data and restricted topology. Any source lacking permission, provenance, mapping or quality is omitted and the related statement marked unsupported. If permission, provenance, calibration, spatial coverage or confounder control is inadequate, classify the result as partial, conflicting or not assessable rather than forcing a conclusion.

Why this is useful: Supports accountable project reporting and reproducibility while protecting research confidentiality, personal data and operational security.

4. Catalogue summary and implementation requirements

The catalogue combines practical research-staff questions with integration-dependent planning, research-grade evidence assessments and restricted safety cases. Readiness identifies the verified sensing, source authority, quality threshold, permissions and safe-failure controls required for a responsible answer.

Questions	80	Eight research-staff-centred sections containing ten questions each.
R1 practical	1	Can be supported with proposed core sensing and basic authoritative spatial or service feeds.
R2 integration-dependent	38	Require booking, access, project, IT, AV, lift, BMS, equipment, maintenance or forecasting integration.
R3 research or restricted	41	Require validated comparison/diagnosis, controlled operational data, or stronger privacy, governance and safety controls.

Recurring system capabilities

- 1

Project and deadline planning
Resolve focused-work blocks, meetings, experiments, computing, visitors, commuting and irregular-hour constraints across project-delivery timescales.
- 2

Room, route and permission reasoning
Use current room purpose, research-staff access, bookings, capacity, specialist authorisation, visitor rules, step-free routes, lift status and closures.
- 3

Environmental, equipment and service evidence
Combine correctly located room sensing with anonymous occupancy, BMS, energy, network, AV, compute, equipment and authorised safety status.
- 4

Forecast, comparison and diagnosis
Predict demand and conditions, build matched baselines, rank alternatives, reconstruct events and separate coincidence from supported diagnosis.
- 5

Evidence quality and provenance
Check room/asset mapping, sampling, clocks, commissioning/calibration, uptime, gaps, drift, source agreement, processing lineage, permissions and direct versus inferred claims.
- 6

Privacy, governance, safety and abstention
Minimise personal and research data, refuse unauthorised access or safety clearance, ask for missing constraints and return not assessable when evidence is inadequate.

QUESTION RECORD STRUCTURE

Each entry can become a system test containing the natural utterance and paraphrases, required entities, minimum room and service evidence, source-precedence rule, quality threshold, expected operation type, permission rule, answerability class, acceptable clarification, uncertainty statement, safe failure, expected answer characteristics and prohibited overclaims.

5. Evidence, permissions and answerability rules

These rules apply across all 80 records and define when the building may answer, qualify, ask for clarification or return not assessable.

Building and room status

The May 2021 plan is a historical spatial reference. Current room purpose, ownership, capacity, research-staff access, bookability, specialist facilities, equipment, entrances, lifts, services and closures require an authoritative current source.

Spatial coverage

A room-level answer requires a correctly located sensor inside that room or a validated served zone. Corridor and adjacent-room data may be labelled context but cannot be silently substituted.

Time window and operation type

Every answer resolves the requested interval and labels its operation as observation, authoritative lookup, calculation, estimate, forecast, diagnosis or recommendation. Preserve observed-at and retrieved-at times; stale evidence is not current status.

Calibration and data quality

Use stated sampling and aggregation, synchronised clocks, commissioning/calibration, uptime, gap and drift flags, maintenance and source agreement. Failed, stale, relocated or unverified points cannot support a precise comparison, forecast or diagnosis.

Research-content minimisation

Do not inspect datasets, code, models, drafts, experiments, presentations, conversations, speech or network traffic. Process only the minimum user-supplied schedule, activity and requirements needed for the request.

Occupancy, sound and privacy

Use anonymous aggregate counts and numerical LAeq or percentile statistics. Do not identify people, retain speech, infer conversation content, track movement or estimate individual productivity.

Environmental interpretation

Direct-reading NDIR CO2 is a ventilation indicator, not complete air quality. Temperature, RH, PM, indicative TVOC, airflow, radiant conditions, outdoor weather and activity may be relevant subject to coverage and quality.

Experiments and specialist spaces

Environmental IoT data cannot authorise research activity or certify safety. Current room function, induction, risk assessment, equipment, compute entitlement and approved procedures come from authoritative systems.

Operational services and permissions

Bookings, access, visitors, lifts, network, AV, compute, maintenance and closures come from authorised feeds. Enforce research-staff permissions and minimise personal, project and schedule data.

Emergency and high-consequence answers

Certified alarms and official communications take precedence. Safety, health, lone-working, ethics and compliance questions require competent authority; missing or conflicting critical evidence requires a limited or not-assessable answer.

FINAL ANSWERABILITY RULE

Answer only when the required variables, current room or served-zone coverage, requested interval, source authority, quality threshold, permissions and safety constraints are adequate. Otherwise ask for clarification, omit unsupported criteria, return confirmed facts with explicit uncertainty, restrict disclosure, or mark the analytical part not assessable.